ai.py - C:\Users\ganes\AppData\Local\Programs\Python\Python313\ai.py (3.13.3)

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```
from collections import deque

def solve():
    start = (3, 3, 1)
    goal = (0, 0, 0)
    q = deque([(start, [])])
    visited = set()

    moves = [(1,0), (2,0), (0,1), (0,2), (1,1)]

    while q:
        state, path = q.popleft()
        m, c, boat = state

        if state == goal:
            for s in path + [state]:
                print(s)
            return

        if state in visited:
            continue
        visited.add(state)

        for dm, dc in moves:
            if boat == 1:
                nm, nc, nb = m-dm, c-dc, 0
            else:
                nm, nc, nb = m+dm, c+dc, 1

            if 0 <= nm <= 3 and 0 <= nc <= 3:
                rm, rc = 3-nm, 3-nc

                if (nm == 0 or nm >= nc) and (rm == 0 or rm >= rc):
                    new = (nm, nc, nb)
                    if new not in visited:
                        q.append((new, path + [state]))

solve()
```

IDLE Shell 3.13.3

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```
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr  8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\ganes\AppData\Local\Programs\Python\Python313\ai.py ====
(3, 3, 1)
(3, 1, 0)
(3, 2, 1)
(3, 0, 0)
(3, 1, 1)
(1, 1, 0)
(2, 2, 1)
(0, 2, 0)
(0, 3, 1)
(0, 1, 0)
(1, 1, 1)
(0, 0, 0)
>>>
```